Affordable Housing Demand (HD): The effective demand for affordable housing.
Affordable Housing Need (AN): The total need for affordable housing.
Affordable Home Building (AB): The number of affordable homes being built.
Affordable Housing Stock (HS): The current stock of affordable housing.
Housebuilding Market (HM): The overall market for housebuilding.
Absorption Rate (AR): The rate at which new housing stock is absorbed by the market.
Planning Value Capture (PVC): The portion of value captured through planning, given as 35%.
Affordable Market Supply (AMS): The supply of affordable housing in the market.
Given the relationships and terms, here's a potential accounting identity:

Monica GPT-4o

Certainly! Here's a potential accounting identity that incorporates the given terms and relationships:

$$\mathrm{HD} + \mathrm{AN} = \mathrm{AMS} + (\mathrm{HS} + \mathrm{AB})$$

Where:

- HD is the effective demand for affordable housing.
- AN is the total need for affordable housing.
- AMS is the supply of affordable housing in the market.
- HS is the current stock of affordable housing.
- AB is the number of affordable homes being built.

This identity suggests that the combined effective demand and total need for affordable housing should be equal to the sum of the affordable market

supply and the total affordable housing stock (both existing and under construction).

If you have any specific constraints or additional relationships you'd like to incorporate, please let me know!

Stock of Rented Housing = 35% England Housing Stock = 23,950,000 Wales Housing Stock 1,395,000

Monica GPT-4o

Given the stock of rented housing is 35%, we can calculate the stock of rented housing for both England and Wales.

Stock of Rented Housing in England

Total Housing Stock in England: 23,950,000

$$\text{Stock of Rented Housing in England} = 0.35 \times 23,950,000$$

$$\text{Stock of Rented Housing in England} = 8,382,500$$

Stock of Rented Housing in Wales

Total Housing Stock in Wales: 1,395,000

$$\text{Stock of Rented Housing in Wales} = 0.35 \times 1,395,000$$

$$\text{Stock of Rented Housing in Wales} = 488,250$$

Summary

- **Stock of Rented Housing in England:** 8,382,500
- **Stock of Rented Housing in Wales:** 488,250

If you need further calculations or additional information, feel free to ask!

housing gap was in excess of one million homes (HG) An estimated 8.4 million people in England are living in an unaffordable, insecure or unsuitable home, according to the

National Housing Federation(8.4m = AN) approx 10m households are in rented accomoidation oif which 84% is either unaffordable , insecure or unsuitable whic is the AN= (.4m households. Net additional New Build homes in 2023 were 235,000 in the past 35 years the highest number buildt was 250,000 in 2019. With an absorption rate for new build homes of o.6 presently 150,000 new homes may be built this year giving 60,000 affordable homes With a Housing Gap in excess of 1 million homes . With affordability getting worse for Home owners , with a continuing increase in the housing Gap the Home@ix formula shows that to Take a ten year view on solving the problem assuming a Private housebuilder contribuition of 50,000 homes coupled to their absorption rate determined out put an additional 50,000 homes need to be built over and above the past 25yr average each year for ten years. Plug those figures into the Home@ix Formula
Affordable Housing Need (AN)
=
HomeMaker Effective Demand (HD)
+
Affordable Home Building (AB)
+
Affordable Housing Stock (HS)
−
Affordable Market Supply (AMS)
Affordable Housing Need (AN)=HomeMaker Effective Deman...
−Affordable Market Supply (AMS)

This identity balances the need for affordable housing with the effective demand, current building efforts, existing stock, and market supply.?

Monica GPT-4o

Let's break down the provided information and plug the figures into the Home@ix formula:

Provided Information:

1. **Affordable Housing Need (AN):** 8.4 million people (or households) living in unaffordable, insecure, or unsuitable homes.
2. **Housing Gap (HG):** In excess of 1 million homes.
3. **Rented Accommodation:** Approximately 10 million households, with 84% (8.4 million) being unaffordable, insecure, or unsuitable.
4. **Net Additional New Build Homes (2023):** 235,000.
5. **Highest Number Built (2019):** 250,000.
6. **Absorption Rate (AR):** 0.6.
7. **Estimated New Homes Built This Year:** 150,000, with 60,000 being affordable homes.
8. **Private Housebuilder Contribution:** 50,000 homes per year for ten years.
9. **Additional Homes Needed:** 50,000 homes over and above the past 25-year average each year for ten years.

Home@ix Formula:

$$\begin{aligned}&\text{Affordable Housing Need (AN)} = \\ &\text{HomeMaker Effective Demand (HD)} + \\ &\text{Affordable Home Building (AB)} + \\ &\text{Affordable Housing Stock (HS)} - \\ &\text{Affordable Market Supply (AMS)}\end{aligned}$$

Calculations:

1. **Affordable Housing Need (AN):** 8.4 million households.
2. **Current Affordable Home Building (AB):** 60,000 homes (this year estimated).
3. **Affordable Housing Stock (HS):** Let's assume it's the current stock of rented housing at 35% of total housing stock in England: $\text{HS} = 0.35 \times 23,950,000 = 8,382,500$
4. **Affordable Market Supply (AMS):** This needs to be determined or given.

Additional Calculations:

- **Total Homes Needed Over Ten Years:** 50,000 (private housebuilder contribution) + 50,000 (additional homes) = 100,000 homes per year.
- **Total Additional Homes Over Ten Years:** 100,000 homes/year $\times$ 10 years = 1,000,000 homes.

Plugging into the Formula:

Given the information, we need to balance the formula. Let's assume the HomeMaker Effective Demand (HD) is the housing gap plus the current demand:

$$\mathrm{HD} = \mathrm{HG} + \mathrm{AN} = 1,000,000 + 8,400,000 = 9,400,000$$

Now, let's plug the values into the formula:

$$8,400,000 = 9,400,000 + 60,000 + 8,382,500 - \mathrm{AMS}$$

Solving for AMS:

$$\mathrm{AMS} = 9,400,000 + 60,000 + 8,382,500 - 8,400,000$$

$$\mathrm{AMS} = 9,442,500 - 8,400,000$$

$$\mathrm{AMS} = 1,042,500$$

Summary:

To balance the Home@ix formula, the Affordable Market Supply (AMS) should be approximately 1,042,500. This calculation shows that the current efforts still leave a significant gap, indicating the need for increased building efforts and policy interventions to address the affordable housing crisis.

If you have more specific details or additional constraints, we can further refine the calculations.

Mortgage Lenders and Administrators Statistics: 2022 Q4
Key findings
The outstanding value of all residential mortgage loans was £1,675.8 billion at the end of 2022 Q4, 3.9% higher than a year

earlier (Table A). footnote[1]

1,675,800,000,000 Value of Outstanding Mortgages on Bank Balance Sheets

Secured against By Savills Figure for Jan 2022
£2,900,000,000,000 £2.9 Trillion of Value

1675.8/2900 = 57.786% LTV

If as some commentators Claim Values Fall by 35% in real terms

The Value of all Mortgaged Property would fall to 1.875 Trillion and the Loan to value ratio on Bank Balance sheets would rise to 88.9% This would have very severe consequences on the rating of Mortgage Debt held by banks as secured collateral for Tier 1 Basel 3 Solvency purposes.

Regarding Falling Prices at the End of 2021 “unmortgaged owner occupiers continued to be the fastest growing housing tenure in 2021, up 12.0% on the year to reach a total of 3.3 trillion”.

53.2% of the UKs Residential Housing Stock in private ownership is not Mortgaged this figure suggests why Prices for Property are much stickier going down than going Up. SHow the Finance Circuit for Private house building linked to Absorption rates and Then Compare the needed Capitall circuit to Build 100,000 Affordable Homes to Meet the Housing Gap, then consider the (.5 m in Housing need ? make the figures explicit in relation to the home@ix Formula as given in the following and above The Home@ix Formula.

HomeMaker Effective Demand=HD
Affordable Housing Need = AN
Affordable Home Building= AB

Affordable Housing Stock =HS

Housebuilding Market =HM
Absorption Rate = AR

Planning Value Capture = 35%

Affordable Market supply = AMS

AMS X AR X PVC = AMS

AN > HD

To create an accounting identity using the provided terms from The Home@ix Formula, we need to relate these terms in a meaningful way. Here's a possible accounting identity based on the given terms:

Affordable Housing Demand (HD): The effective demand for affordable housing.
Affordable Housing Need (AN): The total need for affordable housing.
Affordable Home Building (AB): The number of affordable homes being built.
Affordable Housing Stock (HS): The current stock of affordable housing.
Housebuilding Market (HM): The overall market for housebuilding.
Absorption Rate (AR): The rate at which new housing stock is absorbed by the market.
Planning Value Capture (PVC): The portion of value captured through planning, given as 35%.
Affordable Market Supply (AMS): The supply of affordable housing in the market.
Given the relationships and terms, here’s a potential accounting identity:

Affordable Market Supply (AMS)
×
Absorption Rate (AR)
×
Planning Value Capture (PVC)
=
Affordable Market Supply (AMS)
Affordable Market Supply (AMS)×Absorption Rate (AR)×Plan...

This simplifies to:

AMS
=
AMS
AMS=AMS

This identity holds true as a tautology, but it doesn't provide much insight. Instead, let's relate the terms to form a more informative identity. Considering the need (AN) and the effective demand (HD):

Affordable Housing Need (AN)
>
HomeMaker Effective Demand (HD)
Affordable Housing Need (AN)>HomeMaker Effective Deman...

We also know that:

Affordable Home Building (AB)
+
Affordable Housing Stock (HS)
=
Affordable Housing Supply (AMS)
Affordable Home Building (AB)+Affordable Housing Stock (H...

To incorporate the absorption rate and planning value capture:

AMS
×
AR
×
0.35
=
AMS
AMS×AR×0.35=AMS

Given these, a more comprehensive identity might be:

Affordable Housing Need (AN)
=
HomeMaker Effective Demand (HD)
+
Affordable Home Building (AB)
+
Affordable Housing Stock (HS)
−
Affordable Market Supply (AMS)
Affordable Housing Need (AN)=HomeMaker Effective Deman…
−Affordable Market Supply (AMS)

This identity balances the need for affordable housing with the effective demand, current building efforts, existing stock, and market supply.

Genius
You

Make an accounting Identity from the following defined terms The Home@ix Formula.

HomeMaker Effective Demand = HD
Affordable Housing Need = AN
Affordable Home Building = AB
Affordable Housing Stock = HS
Housebuilding Market = HM
Absorption Rate = AR
Planning Value Capture = 35%
Affordable Market supply = AMS
AMS X AR X PVC = AMS

AN > HD

Monica
To create an accounting identity using the provided terms from The Home@ix Formula, we need to relate these terms in a meaningful way. Here's a possible accounting identity based on the given terms:

* Affordable Housing Demand (HD): The effective demand for affordable housing.
* Affordable Housing Need (AN): The total need for affordable housing.
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* Affordable Housing Stock (HS): The current stock of affordable housing.
* Housebuilding Market (HM): The overall market for housebuilding.
* Absorption Rate (AR): The rate at which new housing stock is absorbed by the market.
* Planning Value Capture (PVC): The portion of value captured through planning, given as 35%.
* Affordable Market Supply (AMS): The supply of affordable housing in the market.

Given the relationships and terms, here's a potential accounting identity:

Affordable Market Supply (AMS) × Absorption Rate (AR) × Planning Value Capture (PVC) = Affordable Market Supply (AMS)

Affordable Market Supply (AMS) × Absorption Rate (AR) × Planning Value Capture (PVC) = Affordable Market Supply (AMS)

This simplifies to:

AMS = AMS
AMS = AMS

This identity holds true as a tautology, but it doesn't provide much insight. Instead, let's relate the terms to form a more informative identity. Considering the need (AN) and the effective demand (HD):

Affordable Housing Need (AN) > HomeMaker Effective Demand (HD)
Affordable Housing Need (AN) > HomeMaker Effective Demand (HD)

We also know that:

Affordable Home Building (AB) + Affordable Housing Stock (HS) = Affordable Housing Supply (AMS)
Affordable Home Building (AB) + Affordable Housing Stock (HS) = Affordable Housing Supply (AMS)

To incorporate the absorption rate and planning value capture:

AMS × AR × 0.35 = AMS

AMS × AR × 0.35 = AMS

Given these, a more comprehensive identity might be:

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + Affordable Home Building (AB) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + Affordable Home Building (AB) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

This identity balances the need for affordable housing with the effective demand, current building efforts, existing stock, and market supply.

You
How does the "Housebuilding Market (HM)" factor into the overall accounting identity of the Home@ix Formula?

Monica
To incorporate the "Housebuilding Market (HM)" into the overall accounting identity of the Home@ix Formula, we need to understand how the broader market for housebuilding interacts with the specific elements of affordable housing. The Housebuilding Market (HM) represents the entire market for building houses, which includes both affordable and non-affordable housing.

Here’s a way to integrate HM into the formula:

* Affordable Housing Demand (HD): The effective demand for affordable housing.
* Affordable Housing Need (AN): The total need for affordable housing.
* Affordable Home Building (AB): The number of affordable

homes being built.
* Affordable Housing Stock (HS): The current stock of affordable housing.
* Housebuilding Market (HM): The overall market for housebuilding.
* Absorption Rate (AR): The rate at which new housing stock is absorbed by the market.
* Planning Value Capture (PVC): The portion of value captured through planning, given as 35%.
* Affordable Market Supply (AMS): The supply of affordable housing in the market.

Given these terms, we need to relate the Housebuilding Market (HM) to the supply and demand for affordable housing. One way to do this is by considering that a portion of the overall housebuilding market (HM) is dedicated to affordable housing (AB).

Here's a proposed accounting identity that includes HM:

Housebuilding Market (HM) × Affordable Housing Proportion (P) = Affordable Home Building (AB)
Housebuilding Market (HM) × Affordable Housing Proportion (P) = Affordable Home Building (AB)

Where *P* is the proportion of the total housebuilding market dedicated to affordable housing.

Now, integrating this into the overall accounting identity, we get:

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P) + Affordable

Housing Stock (HS) - Affordable Market Supply (AMS)

This identity states that the total need for affordable housing (AN) is met by the effective demand (HD), the portion of the housebuilding market dedicated to affordable housing (HM × P), the existing affordable housing stock (HS), minus the affordable market supply (AMS).

To further refine, considering absorption rate (AR) and planning value capture (PVC):

Affordable Market Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)
Affordable Market Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)

Thus, the overall identity becomes:

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

This identity balances the need for affordable housing with effective demand, contributions from the housebuilding market, existing stock, and market supply.

You
How does the Housebuilding Market (HM) influence the overall supply and demand dynamics in the affordable housing sector?

Monica

The Housebuilding Market (HM) significantly influences the overall supply and demand dynamics in the affordable housing sector through several key mechanisms. Understanding these influences can help in formulating policies and strategies to address affordable housing needs effectively. Here's a detailed breakdown:

Influence on Supply

* **Proportion of Affordable Housing (P):**
  * The HM includes both affordable and non-affordable housing. The proportion of HM dedicated to affordable housing (P) directly affects the supply of affordable homes.
  * If a larger portion of HM is allocated to affordable housing, the supply of affordable homes (AB) increases.
* **Construction Activity:**
  * Increased activity in HM generally leads to more construction projects, including affordable housing. This can be influenced by government incentives, subsidies, and regulations that encourage or mandate the inclusion of affordable units in new developments.
* **Economies of Scale:**
  * Larger housebuilding markets can achieve economies of scale, reducing the per-unit cost of construction. This can make it more feasible for developers to build affordable housing.

Influence on Demand

* **Market Dynamics:**
  * The overall health of HM affects housing prices and availability. In a robust HM, increased supply can help stabilize or reduce housing prices, making it easier for people to afford homes.
  * Conversely, in a constrained HM with high demand and limited supply, housing prices can rise, making it more difficult for low-income families to find affordable options.
* **Absorption Rate (AR):**
  * The rate at which new housing stock is absorbed by the

market (AR) affects how quickly new affordable housing becomes occupied. A high AR indicates strong demand and quick occupancy, while a low AR might suggest oversupply or mismatched demand.

Policy and Planning
* **Planning Value Capture (PVC):**
  * Policies that capture a portion of the value created by new developments (e.g., through taxes or fees) can be reinvested into affordable housing projects. PVC at 35% can provide significant funding for affordable housing initiatives.
* **Regulatory Environment:**
  * Zoning laws, building codes, and other regulations within HM can either facilitate or hinder the development of affordable housing. Favorable regulations can streamline the approval process and reduce costs for affordable housing projects.

Integrating HM into the Home@ix Formula
Given these influences, let's refine the accounting identity to reflect the role of HM more clearly:

Affordable Housing Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)

Affordable Housing Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

This refined identity highlights how the overall housebuilding market (HM) influences both the supply (through the proportion of affordable housing, absorption rate, and planning value capture) and the ability to meet the demand. Given the above Identities and formulas, Provide a more comprehensive analysis, Showing Money Supply From Mortgage Lending, Where M3 = .75% made up of new morgage lending and the stock of Money is a similar proportion. The Money Supply is also therefors tied to Absorbtion rates and not just the SUpply of Affordable Housing through S.10 Planning Value Capture at 35%. Show that to Meet Affordable Housing Need , Effective demand has to be placed into the hands of those in Affordable Housing need and that this requires a new circuit of Credit creation, and not an expansion of debt. The New circuit would not be inflationary and could be both Rent controlled and tied to an affordabilty function in line with The Free Housing MArket, Show a term which states that the Free Market has a Maximum limit to the affordable housing provision it can make as Land Value capture due to the absorption rate , The Free Housing Market and Stock of Mortgage debt is fundamental to the moey supply due to financialisation. A new circuiot of Credit creation for the Distinct and discreet Affordable Housing market has to be stated clearly and seperately to Make any advances in solving the problem of Affordability not just in the UK but across the G7, which is a financialisation problem not a planning or geospatial problem.

Certainly! Let's break down the elements and create a comprehensive analysis that incorporates the concepts of money supply, mortgage lending, and the need for a new circuit of credit creation for affordable housing.

Genius

You

Make an accounting Identity from the following defined terms The Home@ix Formula.

- HomeMaker Effective Demand = HD
- Affordable Housing Need = AN
- Affordable Home Building = AB
- Affordable Housing Stock = HS
- Housebuilding Market = HM
- Absorption Rate = AR
- Planning Value Capture = 35%
- Affordable Market Supply = AMS

$$ \text{AMS} \times \text{AR} \times \text{PVC} = \text{AMS} $$

$$ \text{AN} > \text{HD} $$

Monica

To create an accounting identity using the provided terms from The Home@ix Formula, we need to relate these terms in a meaningful way. Here's a possible accounting identity based on the given terms:

- **Affordable Housing Demand (HD)**: The effective demand for affordable housing.
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- **Planning Value Capture (PVC)**: The portion of value captured through planning, given as 35%.
- **Affordable Market Supply (AMS)**: The supply of affordable

housing in the market.

Given the relationships and terms, here’s a potential accounting identity:

$$ \text{Affordable Market Supply (AMS)} \times \text{Absorption Rate (AR)} \times \text{Planning Value Capture (PVC)} = \text{Affordable Market Supply (AMS)} $$

$$ \text{AMS} = \text{AMS} $$

This identity holds true as a tautology, but it doesn't provide much insight. Instead, let's relate the terms to form a more informative identity. Considering the need (AN) and the effective demand (HD):

$$ \text{Affordable Housing Need (AN)} > \text{HomeMaker Effective Demand (HD)} $$

We also know that:

$$ \text{Affordable Home Building (AB)} + \text{Affordable Housing Stock (HS)} = \text{Affordable Housing Supply (AMS)} $$

To incorporate the absorption rate and planning value capture:

$$ \text{AMS} \times \text{AR} \times 0.35 = \text{AMS} $$

Given these, a more comprehensive identity might be:

$$ \text{Affordable Housing Need (AN)} = \text{HomeMaker Effective Demand (HD)} + \text{Affordable Home Building (AB)} + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

This identity balances the need for affordable housing with the

effective demand, current building efforts, existing stock, and market supply.

You
How does the "Housebuilding Market (HM)" factor into the overall accounting identity of the Home@ix Formula?

Monica
To incorporate the "Housebuilding Market (HM)" into the overall accounting identity of the Home@ix Formula, we need to understand how the broader market for housebuilding interacts with the specific elements of affordable housing. The Housebuilding Market (HM) represents the entire market for building houses, which includes both affordable and non-affordable housing.

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way to do this is by considering that a portion of the overall housebuilding market (HM) is dedicated to affordable housing (AB).

Here's a proposed accounting identity that includes HM:

$$ \text{Housebuilding Market (HM)} \times \text{Affordable Housing Proportion (P)} = \text{Affordable Home Building (AB)} $$

Where P is the proportion of the total housebuilding market dedicated to affordable housing.

Now, integrating this into the overall accounting identity, we get:

$$ \text{Affordable Housing Need (AN)} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

This identity states that the total need for affordable housing (AN) is met by the effective demand (HD), the portion of the housebuilding market dedicated to affordable housing (HM × P), the existing affordable housing stock (HS), minus the affordable market supply (AMS).

To further refine, considering absorption rate (AR) and planning value capture (PVC):

$$ \text{Affordable Market Supply (AMS)} = (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} $$

Thus, the overall identity becomes:

$$ \text{Affordable Housing Need (AN)} = \text{HomeMaker}$$

Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} \]

This identity balances the need for affordable housing with effective demand, contributions from the housebuilding market, existing stock, and market supply.

You
How does the Housebuilding Market (HM) influence the overall supply and demand dynamics in the affordable housing sector?

Monica
The Housebuilding Market (HM) significantly influences the overall supply and demand dynamics in the affordable housing sector through several key mechanisms. Understanding these influences can help in formulating policies and strategies to address affordable housing needs effectively. Here's a detailed breakdown:

Influence on Supply
- **Proportion of Affordable Housing (P)**:
  - The HM includes both affordable and non-affordable housing. The proportion of HM dedicated to affordable housing (P) directly affects the supply of affordable homes.
  - If a larger portion of HM is allocated to affordable housing, the supply of affordable homes (AB) increases.
- **Construction Activity**:
  - Increased activity in HM generally leads to more construction projects, including affordable housing. This can be influenced by government incentives, subsidies, and regulations that encourage or mandate the inclusion of affordable units in new developments.
- **Economies of Scale**:
  - Larger housebuilding markets can achieve economies of scale, reducing the per-unit cost of construction. This can make it more feasible for developers to build affordable housing.

Influence on Demand
- **Market Dynamics**:
  - The overall health of HM affects housing prices and availability. In a robust HM, increased supply can help stabilize or reduce housing prices, making it easier for people to afford homes.
  - Conversely, in a constrained HM with high demand and limited supply, housing prices can rise, making it more difficult for low-income families to find affordable options.
- **Absorption Rate (AR)**:
  - The rate at which new housing stock is absorbed by the market (AR) affects how quickly new affordable housing becomes occupied. A high AR indicates strong demand and quick occupancy, while a low AR might suggest oversupply or mismatched demand.

Policy and Planning
- **Planning Value Capture (PVC)**:
  - Policies that capture a portion of the value created by new developments (e.g., through taxes or fees) can be reinvested into affordable housing projects. PVC at 35% can provide significant funding for affordable housing initiatives.
- **Regulatory Environment**:
  - Zoning laws, building codes, and other regulations within HM can either facilitate or hinder the development of affordable housing. Favorable regulations can streamline the approval process and reduce costs for affordable housing projects.

Integrating HM into the Home@ix Formula
Given these influences, let’s refine the accounting identity to reflect the role of HM more clearly:

$$ \text{Affordable Housing Supply (AMS)} = (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} $$

\[\text{Affordable Housing Need (AN)} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} \]

This refined identity highlights how the overall housebuilding market (HM) influences both the supply (through the proportion of affordable housing, absorption rate, and planning value capture) and the ability to meet the demand for affordable housing.

Summary
- **Supply Side**: HM affects the number of affordable homes built (AB) through construction activity, economies of scale, and policies encouraging affordable housing.
- **Demand Side**: HM impacts housing prices and availability, influencing the effective demand for affordable housing (HD).
- **Policy and Planning**: Regulatory frameworks and value capture mechanisms within HM can provide resources and incentives for affordable housing development.

By understanding and leveraging these dynamics, policymakers and planners can better address the affordable housing crisis.

You
Incorporate this statement of The Problem of absorption rates stated in the Letwin Review into the Home@ix formula.

First, the problem. The analysis in the 2018 Letwin Independent Review of Build Out Rates still holds good today. He correctly identified that build-out rates for large sites are directly linked to the rate at which new homes can (in the view of the developer) be sold without reducing the local market price. The homogeneity of the types and tenures of homes on offer and the limits on the rate at which the market will absorb

them are the fundamental drivers of the slow rate of build-out. It's not a matter of low demand, it's saturation of the market for houses that look the same and are priced the same. For those interested, the article by Cole & Archer on the financialisation of housing production shows that after the Global Financial Crash (GFC) of 2008 UK volume housebuilders focused on increasing margins per home sold over increasing volume built, increasing volumes by 70% from 2010 to 2017 but increasing profits before tax by 703% during the same period. Part of Letwin's solution was to offer homes of 'varying types, designs and tenures including a high proportion of affordable housing'. He also proposed 'decoupling' affordable homes from full market rate homes, bringing forward all affordable provision to meet demand.

The demand for affordable homes to rent is so great that even an ambitious proposal to build 2,000 homes over 5 years will barely dent the supply. That's why the company has a longer-term plan to scale the Home@ix system.

The car industry's agency model, which has undergone an adaptation to a direct-to-customer hybrid approach, provides the key. This model is now available to house-builders through the Home@ix dashboard, which is a direct-to-consumer model. The dashboard can cater to both fleet buyers (Housing Associations, Registered Social Landlords, or volume housebuilders needing to fulfil their affordable housing condition) and on a smaller scale direct to individual homemakers. The dashboard is agnostic about its customers and can therefore serve both equally well.

The existing sales outlet-based model (the staffed on-site show home) is ill-suited in the shift to buy-to-rent (BTR) mix. There's a strong need for a tenant-based Customer Relationship Management (CRM) forward model. Contractors and sub-contractors are playing an important role in this change. The use of Modern Methods of Construction (MMC) is becoming

more prevalent across the sector. The volume housebuilders remain wedded to their traditional methods of construction and their traditional sales methods.

By leveraging technology and carefully evaluating local market conditions, home builders and BTR investors can establish strong regional sales networks that drive absorption rates and increase sales. Launching with this retail/distribution model delivers exceptional information to the homemaker. Fully ESG compliant, carbon-neutral off-site building manufacture system based on the Verrym@ix DFMA closed panel off-site system.

Monica
Incorporating the insights from the Letwin Review into the Home@ix Formula involves addressing the issues related to absorption rates, market saturation, and the need for diverse housing types and tenures. The Letwin Review emphasizes that build-out rates are linked to the ability to sell new homes without depressing local market prices, which is influenced by the homogeneity of housing types and tenures.

Here's how we can integrate these insights into the Home@ix Formula:

Key Points from the Letwin Review
- **Absorption Rates (AR)**: Linked to the rate at which new homes can be sold without reducing local market prices.
- **Homogeneity**: Homogeneous housing types and tenures slow down build-out rates.
- **Diverse Housing Types and Tenures**: Increasing the variety of housing can improve absorption rates.
- **Decoupling Affordable Homes**: Bringing forward affordable housing provision can meet demand more effectively.
- **Direct-to-Consumer Model**: Leveraging technology to improve sales and absorption rates.

Updated Home@ix Formula
To reflect these points, we need to adjust the formula to account for the impact of diverse housing types and the direct-to-consumer model on absorption rates.

Affordable Housing Supply (AMS)
$$ \text{AMS} = (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} $$

Where:
- **P**: Proportion of HM dedicated to affordable housing.
- **AR**: Absorption rate, now reflecting the diversity of housing types and tenures.
- **PVC**: Planning value capture.
- **HS**: Existing affordable housing stock.

Affordable Housing Need (AN)
$$ \text{AN} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

Incorporating the Letwin Review Insights

1. **Diverse Housing Types and Tenures**:
   - Introduce a factor D representing the diversity of housing types and tenures. Higher diversity improves absorption rates.
   - Modify AR to include D: $AR \times D$.

2. **Direct-to-Consumer Model**:
   - Introduce a factor T representing the effectiveness of the direct-to-consumer model in improving sales and absorption rates.
   - Modify AR to include T: $AR \times T$.

3. **Decoupling Affordable Homes**:
   - Ensure that the formula accounts for the immediate provision of affordable homes, represented by a factor F for fast-tracking affordable housing.

Final Updated Formula

Affordable Housing Supply (AMS)
$$\text{AMS} = (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC) + \text{Affordable Housing Stock (HS)}$$

Affordable Housing Need (AN)
$$\text{AN} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC \times F) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)}$$

Explanation

- **D**: Diversity of housing types and tenures. Higher diversity improves absorption rates by reducing market saturation.
- **T**: Effectiveness of the direct-to-consumer model. This factor reflects how well the Home@ix dashboard and CRM system enhance sales and absorption rates.
- **F**: Fast-tracking affordable homes. This factor ensures that affordable homes are brought forward to meet immediate demand.

Summary
By incorporating the Letwin Review's insights, the updated Home@ix Formula now better reflects the complexities of the housing market, including the importance of diverse housing types, the role of technology in improving sales, and the need to fast-track affordable housing. This comprehensive approach aims to address both supply and demand dynamics more

effectively, ensuring that the affordable housing sector can meet the growing needs of the population.

You

Provide a more comprehensive analysis, showing money supply from mortgage lending, where M3 = 0.75% made up of new mortgage lending and the stock of money is a similar proportion. The money supply is also therefore tied to absorption rates and not just the supply of affordable housing through S.10 Planning Value Capture at 35%. Show that to meet affordable housing needs, effective demand has to be placed into the hands of those in affordable housing need and that this requires a new circuit of credit creation, and not an expansion of debt. The new circuit would not be inflationary and could be both rent-controlled and tied to an affordability function in line with the free housing market. Show a term which states that the free market has a maximum limit to the affordable housing provision it can make as land value capture due to the absorption rate. The free housing market and stock of mortgage debt are fundamental to the money supply due to financialisation. A new circuit of credit creation for the distinct and discreet affordable housing market has to be stated clearly and separately to make any advances in solving the problem of affordability not just in the UK but across the G7, which is a financialisation problem, not a planning or geospatial problem.

Monica

To provide a more comprehensive analysis, we need to incorporate the relationship between the money supply, mortgage lending, absorption rates, and the supply of affordable housing. We will also address the need for a new circuit of credit creation specifically for affordable housing.

Key Concepts

- **Money Supply (M3)**: Represents the total stock of money in the economy, including new mortgage lending.
- **Absorption Rates (AR)**: The rate at which new housing

stock is absorbed by the market.
- **Planning Value Capture (PVC)**: The portion of value captured through planning, given as 35%.
- **Effective Demand (HD)**: The demand for affordable housing that can be met with available financial resources.
- **Affordable Housing Need (AN)**: The total need for affordable housing.
- **New Circuit of Credit Creation**: A mechanism to provide credit for affordable housing without expanding overall debt.

Analysis

Money Supply and Mortgage Lending
The money supply (M3) is

Monica GPT-4o

The money supply (M3) is significantly influenced by the volume of mortgage lending. Given that M3 is composed of 0.75% new mortgage lending and the stock of money is also a similar proportion, we can see how closely tied the housing market is to the overall financial system.

Money Supply from Mortgage Lending

- **Outstanding Value of Residential Mortgage Loans**: £1,675.8 billion.
- **Total Value of Mortgaged Property (Pre-Potential Fall)**: £2.9 trillion.
- **Loan-to-Value Ratio (LTV)**: 57.786%.
- **Potential Value of Mortgaged Property (Post-35% Fall)**: £1.875 trillion.
- **New Mortgage Lending as Part of M3**: 0.75%.

Impact of Absorption Rates

Absorption rates (AR) directly impact how quickly new housing stock, including affordable housing, is taken up by the market. A higher AR indicates that homes are sold and occupied quickly, which can stabilize or

increase housing prices. Conversely, a lower AR can indicate oversupply or mismatched demand, leading to slower sales and potential price decreases.

Planning Value Capture (PVC)

Planning value capture at 35% represents the portion of the value created by new developments that can be reinvested into affordable housing projects. This mechanism can provide significant funding for affordable housing initiatives, but it alone is not sufficient to meet the entire affordable housing need.

Effective Demand and Affordable Housing Need

To meet affordable housing needs, effective demand (HD) must be placed into the hands of those in need. This requires a new circuit of credit creation that is distinct from traditional mortgage lending and does not expand overall debt levels.

New Circuit of Credit Creation

A new circuit of credit creation for affordable housing can be designed to be non-inflationary, rent-controlled, and tied to an affordability function. This circuit would operate separately from the traditional housing finance market, ensuring that funds are specifically allocated to affordable housing projects.

Incorporating the Free Market Limitations

The free market has a maximum limit to the affordable housing it can provide due to land value capture and absorption rates. The financialisation of the housing market means that the stock of mortgage debt is fundamental to the money supply. Therefore, a new credit creation circuit for affordable housing must be clearly stated and separated from traditional market mechanisms.

Comprehensive Accounting Identity

To integrate these elements, we can refine the Home@ix Formula to include the impact of money supply, absorption rates, and the need for a new credit circuit.

Affordable Housing Supply (AMS)

$$\text{AMS} = (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC) + \text{Affordable Housing Stock (HS)}$$

Affordable Housing Need (AN)

$$\text{AN} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC \times F) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)}$$

New Circuit of Credit Creation

To represent the new circuit of credit creation (NCC), we introduce a term C which represents the credit specifically allocated to affordable housing without expanding overall debt.

Updated Affordable Housing Need (AN)

$$\text{AN} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC \times F) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} + \text{New Circuit of Credit Creation (NCC)}$$

Explanation

- D: Diversity of housing types and tenures.
- T: Effectiveness of the direct-to-consumer model.
- F: Fast-tracking affordable homes.
- NCC: New Circuit of Credit Creation, representing non-inflationary, rent-controlled credit for affordable housing.

Summary

This comprehensive analysis shows that to meet affordable housing needs, effective demand must be placed into the hands of those in need through a new circuit of credit creation. This circuit should be distinct from traditional mortgage lending and should not expand overall debt levels. By incorporating the limitations of the free market and the financialisation of

the housing market, we can better address the affordable housing crisis not just in the UK but across the G7. This approach recognizes that the problem is primarily financial, not merely planning or geospatial.

The Home@ix Formula.

HomeMaker Effective Demand=HD
Affordable Housing Need = AN
Affordable Home Building= AB

Affordable Housing Stock =HS

Housebuilding Market =HM
Absorption Rate = AR

Planning Value Capture = 35%

Affordable Market supply = AMS

AMS X AR X PVC = AMS

AN > HD

To create an accounting identity using the provided terms from The Home@ix Formula, we need to relate these terms in a meaningful way. Here's a possible accounting identity based on the given terms:

Affordable Housing Demand (HD): The effective demand for affordable housing.
Affordable Housing Need (AN): The total need for affordable housing.
Affordable Home Building (AB): The number of affordable homes being built.
Affordable Housing Stock (HS): The current stock of affordable housing.

Housebuilding Market (HM): The overall market for housebuilding.
Absorption Rate (AR): The rate at which new housing stock is absorbed by the market.
Planning Value Capture (PVC): The portion of value captured through planning, given as 35%.
Affordable Market Supply (AMS): The supply of affordable housing in the market.
Given the relationships and terms, here's a potential accounting identity:

Affordable Market Supply (AMS)
×
Absorption Rate (AR)
×
Planning Value Capture (PVC)
=
Affordable Market Supply (AMS)
Affordable Market Supply (AMS)×Absorption Rate (AR)×Plan...

This simplifies to:

AMS
=
AMS
AMS=AMS

This identity holds true as a tautology, but it doesn't provide much insight. Instead, let's relate the terms to form a more informative identity. Considering the need (AN) and the effective demand (HD):

Affordable Housing Need (AN)
>
HomeMaker Effective Demand (HD)
Affordable Housing Need (AN)>HomeMaker Effective Deman...

We also know that:

Affordable Home Building (AB)
+
Affordable Housing Stock (HS)
=
Affordable Housing Supply (AMS)
Affordable Home Building (AB)+Affordable Housing Stock (H…

To incorporate the absorption rate and planning value capture:

AMS
×
AR
×
0.35
=
AMS
AMS×AR×0.35=AMS

Given these, a more comprehensive identity might be:

Affordable Housing Need (AN)
=
HomeMaker Effective Demand (HD)
+
Affordable Home Building (AB)
+
Affordable Housing Stock (HS)
−
Affordable Market Supply (AMS)
Affordable Housing Need (AN)=HomeMaker Effective Deman…
−Affordable Market Supply (AMS)

This identity balances the need for affordable housing with the effective demand, current building efforts, existing stock, and market supply.

Genius

You

Make an accounting Identity from the following defined terms The Home@ix Formula.

HomeMaker Effective Demand = HD
Affordable Housing Need = AN
Affordable Home Building = AB
Affordable Housing Stock = HS
Housebuilding Market = HM
Absorption Rate = AR
Planning Value Capture = 35%
Affordable Market supply = AMS
AMS X AR X PVC = AMS

AN > HD

Monica

To create an accounting identity using the provided terms from The Home@ix Formula, we need to relate these terms in a meaningful way. Here's a possible accounting identity based on the given terms:

* Affordable Housing Demand (HD): The effective demand for affordable housing.
* Affordable Housing Need (AN): The total need for affordable housing.
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* Affordable Housing Stock (HS): The current stock of affordable housing.
* Housebuilding Market (HM): The overall market for housebuilding.
* Absorption Rate (AR): The rate at which new housing stock is absorbed by the market.
* Planning Value Capture (PVC): The portion of value captured through planning, given as 35%.
* Affordable Market Supply (AMS): The supply of affordable housing in the market.

Given the relationships and terms, here’s a potential accounting identity:

Affordable Market Supply (AMS) × Absorption Rate (AR) × Planning Value Capture (PVC) = Affordable Market Supply (AMS)

Affordable Market Supply (AMS) × Absorption Rate (AR) × Planning Value Capture (PVC) = Affordable Market Supply (AMS)

This simplifies to:

AMS = AMS
AMS = AMS

This identity holds true as a tautology, but it doesn't provide much insight. Instead, let's relate the terms to form a more informative identity. Considering the need (AN) and the effective demand (HD):

Affordable Housing Need (AN) > HomeMaker Effective Demand (HD)
Affordable Housing Need (AN) > HomeMaker Effective Demand (HD)

We also know that:

Affordable Home Building (AB) + Affordable Housing Stock (HS) = Affordable Housing Supply (AMS)
Affordable Home Building (AB) + Affordable Housing Stock (HS) = Affordable Housing Supply (AMS)

To incorporate the absorption rate and planning value capture:

AMS × AR × 0.35 = AMS
AMS × AR × 0.35 = AMS

Given these, a more comprehensive identity might be:

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + Affordable Home Building (AB) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + Affordable Home Building (AB) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

This identity balances the need for affordable housing with the effective demand, current building efforts, existing stock, and market supply.

You
How does the "Housebuilding Market (HM)" factor into the overall accounting identity of the Home@ix Formula?

Monica
To incorporate the "Housebuilding Market (HM)" into the overall accounting identity of the Home@ix Formula, we need to understand how the broader market for housebuilding interacts with the specific elements of affordable housing. The Housebuilding Market (HM) represents the entire market for

building houses, which includes both affordable and non-affordable housing.

Here's a way to integrate HM into the formula:

* Affordable Housing Demand (HD): The effective demand for affordable housing.
* Affordable Housing Need (AN): The total need for affordable housing.
* Affordable Home Building (AB): The number of affordable homes being built.
* Affordable Housing Stock (HS): The current stock of affordable housing.
* Housebuilding Market (HM): The overall market for housebuilding.
* Absorption Rate (AR): The rate at which new housing stock is absorbed by the market.
* Planning Value Capture (PVC): The portion of value captured through planning, given as 35%.
* Affordable Market Supply (AMS): The supply of affordable housing in the market.

Given these terms, we need to relate the Housebuilding Market (HM) to the supply and demand for affordable housing. One way to do this is by considering that a portion of the overall housebuilding market (HM) is dedicated to affordable housing (AB).

Here's a proposed accounting identity that includes HM:

Housebuilding Market (HM) × Affordable Housing Proportion (P) = Affordable Home Building (AB)
Housebuilding Market (HM) × Affordable Housing Proportion (P) = Affordable Home Building (AB)

Where *P* is the proportion of the total housebuilding market dedicated to affordable housing.

Now, integrating this into the overall accounting identity, we get:

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

This identity states that the total need for affordable housing (AN) is met by the effective demand (HD), the portion of the housebuilding market dedicated to affordable housing (HM × P), the existing affordable housing stock (HS), minus the affordable market supply (AMS).

To further refine, considering absorption rate (AR) and planning value capture (PVC):

Affordable Market Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)
Affordable Market Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)

Thus, the overall identity becomes:

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

This identity balances the need for affordable housing with effective demand, contributions from the housebuilding market, existing stock, and market supply.

You
How does the Housebuilding Market (HM) influence the overall supply and demand dynamics in the affordable housing sector?

Monica
The Housebuilding Market (HM) significantly influences the overall supply and demand dynamics in the affordable housing sector through several key mechanisms. Understanding these influences can help in formulating policies and strategies to address affordable housing needs effectively. Here's a detailed breakdown:

Influence on Supply
* **Proportion of Affordable Housing (P):**
  * The HM includes both affordable and non-affordable housing. The proportion of HM dedicated to affordable housing (P) directly affects the supply of affordable homes.
  * If a larger portion of HM is allocated to affordable housing, the supply of affordable homes (AB) increases.
* **Construction Activity:**
  * Increased activity in HM generally leads to more construction projects, including affordable housing. This can be influenced by government incentives, subsidies, and regulations that encourage or mandate the inclusion of affordable units in new developments.
* **Economies of Scale:**
  * Larger housebuilding markets can achieve economies of scale, reducing the per-unit cost of construction. This can make it more feasible for developers to build affordable housing.

Influence on Demand

* **Market Dynamics:**
  * The overall health of HM affects housing prices and availability. In a robust HM, increased supply can help stabilize or reduce housing prices, making it easier for people to afford homes.
  * Conversely, in a constrained HM with high demand and limited supply, housing prices can rise, making it more difficult for low-income families to find affordable options.
* **Absorption Rate (AR):**
  * The rate at which new housing stock is absorbed by the market (AR) affects how quickly new affordable housing becomes occupied. A high AR indicates strong demand and quick occupancy, while a low AR might suggest oversupply or mismatched demand.

Policy and Planning
* **Planning Value Capture (PVC):**
  * Policies that capture a portion of the value created by new developments (e.g., through taxes or fees) can be reinvested into affordable housing projects. PVC at 35% can provide significant funding for affordable housing initiatives.
* **Regulatory Environment:**
  * Zoning laws, building codes, and other regulations within HM can either facilitate or hinder the development of affordable housing. Favorable regulations can streamline the approval process and reduce costs for affordable housing projects.

Integrating HM into the Home@ix Formula
Given these influences, let's refine the accounting identity to reflect the role of HM more clearly:

Affordable Housing Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)

Affordable Housing Supply (AMS) = (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

Affordable Housing Need (AN) = HomeMaker Effective Demand (HD) + (Housebuilding Market (HM) × P × AR × PVC) + Affordable Housing Stock (HS) - Affordable Market Supply (AMS)

This refined identity highlights how the overall housebuilding market (HM) influences both the supply (through the proportion of affordable housing, absorption rate, and planning value capture) and the ability to meet the demand. Given the above Identities and formulas, Provide a more comprehensive analysis, Showing Money Supply From Mortgage Lending, Where M3 = .75% made up of new morgage lending and the stock of Money is a similar proportion. The Money Supply is also therefors tied to Absorbtion rates and not just the SUpply of Affordable Housing through S.10 Planning Value Capture at 35%. Show that to Meet Affordable Housing Need , Effective demand has to be placed into the hands of those in Affordable Housing need and that this requires a new circuit of Credit creation, and not an expansion of debt. The New circuit would not be inflationary and could be both Rent controlled and tied to an affordabilty function in line with The Free Housing MArket, Show a term which states that the Free Market has a Maximum limit to the affordable housing provision it can make as Land Value capture due to the absorption rate , The Free Housing Market and Stock of Mortgage debt is fundamental to the moey supply due to financialisation. A new circuiot of Credit creation for the Distinct and discreet Affordable Housing market has to be stated clearly and seperately to Make any advances in solving the problem of Affordability not just in the UK but across the G7, which is a financialisation problem not a planning or

geospatial problem.

Certainly! Let's break down the elements and create a comprehensive analysis that incorporates the concepts of money supply, mortgage lending, and the need for a new circuit of credit creation for affordable housing.

Genius

You
Make an accounting Identity from the following defined terms The Home@ix Formula.

- HomeMaker Effective Demand = HD
- Affordable Housing Need = AN
- Affordable Home Building = AB
- Affordable Housing Stock = HS
- Housebuilding Market = HM
- Absorption Rate = AR
- Planning Value Capture = 35%
- Affordable Market Supply = AMS

$$ \text{AMS} \times \text{AR} \times \text{PVC} = \text{AMS} $$

$$ \text{AN} > \text{HD} $$

Monica
To create an accounting identity using the provided terms from The Home@ix Formula, we need to relate these terms in a meaningful way. Here's a possible accounting identity based on the given terms:

- **Affordable Housing Demand (HD)**: The effective demand for affordable housing.
- **Affordable Housing Need (AN)**: The total need for affordable housing.
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homes being built.
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- **Absorption Rate (AR)**: The rate at which new housing stock is absorbed by the market.
- **Planning Value Capture (PVC)**: The portion of value captured through planning, given as 35%.
- **Affordable Market Supply (AMS)**: The supply of affordable housing in the market.

Given the relationships and terms, here’s a potential accounting identity:

$$ \text{Affordable Market Supply (AMS)} \times \text{Absorption Rate (AR)} \times \text{Planning Value Capture (PVC)} = \text{Affordable Market Supply (AMS)} $$

$$ \text{AMS} = \text{AMS} $$

This identity holds true as a tautology, but it doesn't provide much insight. Instead, let's relate the terms to form a more informative identity. Considering the need (AN) and the effective demand (HD):

$$ \text{Affordable Housing Need (AN)} > \text{HomeMaker Effective Demand (HD)} $$

We also know that:

$$ \text{Affordable Home Building (AB)} + \text{Affordable Housing Stock (HS)} = \text{Affordable Housing Supply (AMS)} $$

To incorporate the absorption rate and planning value capture:

$$ \text{AMS} \times \text{AR} \times 0.35 = \text{AMS} $$

Given these, a more comprehensive identity might be:

$$ \text{Affordable Housing Need (AN)} = \text{HomeMaker Effective Demand (HD)} + \text{Affordable Home Building (AB)} + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

This identity balances the need for affordable housing with the effective demand, current building efforts, existing stock, and market supply.

You
How does the "Housebuilding Market (HM)" factor into the overall accounting identity of the Home@ix Formula?

Monica
To incorporate the "Housebuilding Market (HM)" into the overall accounting identity of the Home@ix Formula, we need to understand how the broader market for housebuilding interacts with the specific elements of affordable housing. The Housebuilding Market (HM) represents the entire market for building houses, which includes both affordable and non-affordable housing.

Here’s a way to integrate HM into the formula:

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Given these terms, we need to relate the Housebuilding Market (HM) to the supply and demand for affordable housing. One way to do this is by considering that a portion of the overall housebuilding market (HM) is dedicated to affordable housing (AB).

Here's a proposed accounting identity that includes HM:

$$ \text{Housebuilding Market (HM)} \times \text{Affordable Housing Proportion (P)} = \text{Affordable Home Building (AB)} $$

Where P is the proportion of the total housebuilding market dedicated to affordable housing.

Now, integrating this into the overall accounting identity, we get:

$$ \text{Affordable Housing Need (AN)} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

This identity states that the total need for affordable housing (AN) is met by the effective demand (HD), the portion of the housebuilding market dedicated to affordable housing (HM × P), the existing affordable housing stock (HS), minus the affordable market supply (AMS).

To further refine, considering absorption rate (AR) and planning value capture (PVC):

$$ \text{Affordable Market Supply (AMS)} = (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} $$

Thus, the overall identity becomes:

$$ \text{Affordable Housing Need (AN)} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

This identity balances the need for affordable housing with effective demand, contributions from the housebuilding market, existing stock, and market supply.

You
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Monica
The Housebuilding Market (HM) significantly influences the overall supply and demand dynamics in the affordable housing sector through several key mechanisms. Understanding these influences can help in formulating policies and strategies to address affordable housing needs effectively. Here's a detailed breakdown:

Influence on Supply
- **Proportion of Affordable Housing (P)**:
  - The HM includes both affordable and non-affordable housing. The proportion of HM dedicated to affordable housing (P) directly affects the supply of affordable homes.
  - If a larger portion of HM is allocated to affordable housing, the supply of affordable homes (AB) increases.

- **Construction Activity**:
  - Increased activity in HM generally leads to more construction projects, including affordable housing. This can be influenced by government incentives, subsidies, and regulations that encourage or mandate the inclusion of affordable units in new developments.
- **Economies of Scale**:
  - Larger housebuilding markets can achieve economies of scale, reducing the per-unit cost of construction. This can make it more feasible for developers to build affordable housing.

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- **Market Dynamics**:
  - The overall health of HM affects housing prices and availability. In a robust HM, increased supply can help stabilize or reduce housing prices, making it easier for people to afford homes.
  - Conversely, in a constrained HM with high demand and limited supply, housing prices can rise, making it more difficult for low-income families to find affordable options.
- **Absorption Rate (AR)**:
  - The rate at which new housing stock is absorbed by the market (AR) affects how quickly new affordable housing becomes occupied. A high AR indicates strong demand and quick occupancy, while a low AR might suggest oversupply or mismatched demand.

Policy and Planning
- **Planning Value Capture (PVC)**:
  - Policies that capture a portion of the value created by new developments (e.g., through taxes or fees) can be reinvested into affordable housing projects. PVC at 35% can provide significant funding for affordable housing initiatives.
- **Regulatory Environment**:
  - Zoning laws, building codes, and other regulations within HM can either facilitate or hinder the development of affordable housing. Favorable regulations can streamline the

approval process and reduce costs for affordable housing projects.

Integrating HM into the Home@ix Formula
Given these influences, let's refine the accounting identity to reflect the role of HM more clearly:

$$ \text{Affordable Housing Supply (AMS)} = (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} $$

$$ \text{Affordable Housing Need (AN)} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

This refined identity highlights how the overall housebuilding market (HM) influences both the supply (through the proportion of affordable housing, absorption rate, and planning value capture) and the ability to meet the demand for affordable housing.

Summary
- **Supply Side**: HM affects the number of affordable homes built (AB) through construction activity, economies of scale, and policies encouraging affordable housing.
- **Demand Side**: HM impacts housing prices and availability, influencing the effective demand for affordable housing (HD).
- **Policy and Planning**: Regulatory frameworks and value capture mechanisms within HM can provide resources and incentives for affordable housing development.

By understanding and leveraging these dynamics, policymakers and planners can better address the affordable housing crisis.

You

Incorporate this statement of The Problem of absorption rates stated in the Letwin Review into the Home@ix formula.

First, the problem. The analysis in the 2018 Letwin Independent Review of Build Out Rates still holds good today. He correctly identified that build-out rates for large sites are directly linked to the rate at which new homes can (in the view of the developer) be sold without reducing the local market price. The homogeneity of the types and tenures of homes on offer and the limits on the rate at which the market will absorb them are the fundamental drivers of the slow rate of build-out. It's not a matter of low demand, it's saturation of the market for houses that look the same and are priced the same. For those interested, the article by Cole & Archer on the financialisation of housing production shows that after the Global Financial Crash (GFC) of 2008 UK volume housebuilders focused on increasing margins per home sold over increasing volume built, increasing volumes by 70% from 2010 to 2017 but increasing profits before tax by 703% during the same period. Part of Letwin's solution was to offer homes of 'varying types, designs and tenures including a high proportion of affordable housing'. He also proposed 'decoupling' affordable homes from full market rate homes, bringing forward all affordable provision to meet demand.

The demand for affordable homes to rent is so great that even an ambitious proposal to build 2,000 homes over 5 years will barely dent the supply. That's why the company has a longer-term plan to scale the Home@ix system.

The car industry's agency model, which has undergone an adaptation to a direct-to-customer hybrid approach, provides the key. This model is now available to house-builders through the Home@ix dashboard, which is a direct-to-consumer model. The dashboard can cater to both fleet buyers (Housing Associations, Registered Social Landlords, or volume housebuilders needing to fulfil their affordable housing

condition) and on a smaller scale direct to individual homemakers. The dashboard is agnostic about its customers and can therefore serve both equally well.

The existing sales outlet-based model (the staffed on-site show home) is ill-suited in the shift to buy-to-rent (BTR) mix. There's a strong need for a tenant-based Customer Relationship Management (CRM) forward model. Contractors and sub-contractors are playing an important role in this change. The use of Modern Methods of Construction (MMC) is becoming more prevalent across the sector. The volume housebuilders remain wedded to their traditional methods of construction and their traditional sales methods.

By leveraging technology and carefully evaluating local market conditions, home builders and BTR investors can establish strong regional sales networks that drive absorption rates and increase sales. Launching with this retail/distribution model delivers exceptional information to the homemaker. Fully ESG compliant, carbon-neutral off-site building manufacture system based on the Verrym@ix DFMA closed panel off-site system.

Monica
Incorporating the insights from the Letwin Review into the Home@ix Formula involves addressing the issues related to absorption rates, market saturation, and the need for diverse housing types and tenures. The Letwin Review emphasizes that build-out rates are linked to the ability to sell new homes without depressing local market prices, which is influenced by the homogeneity of housing types and tenures.

Here's how we can integrate these insights into the Home@ix Formula:

Key Points from the Letwin Review
- **Absorption Rates (AR)**: Linked to the rate at which new homes can be sold without reducing local market prices.

- **Homogeneity**: Homogeneous housing types and tenures slow down build-out rates.
- **Diverse Housing Types and Tenures**: Increasing the variety of housing can improve absorption rates.
- **Decoupling Affordable Homes**: Bringing forward affordable housing provision can meet demand more effectively.
- **Direct-to-Consumer Model**: Leveraging technology to improve sales and absorption rates.

Updated Home@ix Formula
To reflect these points, we need to adjust the formula to account for the impact of diverse housing types and the direct-to-consumer model on absorption rates.

Affordable Housing Supply (AMS)
$$ \text{AMS} = (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} $$

Where:
- **P**: Proportion of HM dedicated to affordable housing.
- **AR**: Absorption rate, now reflecting the diversity of housing types and tenures.
- **PVC**: Planning value capture.
- **HS**: Existing affordable housing stock.

Affordable Housing Need (AN)
$$ \text{AN} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times PVC) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

Incorporating the Letwin Review Insights

1. **Diverse Housing Types and Tenures**:
   - Introduce a factor D representing the diversity of

housing types and tenures. Higher diversity improves absorption rates.
  - Modify AR to include D: $AR \times D$.

2. **Direct-to-Consumer Model**:
  - Introduce a factor T representing the effectiveness of the direct-to-consumer model in improving sales and absorption rates.
  - Modify AR to include T: $AR \times T$.

3. **Decoupling Affordable Homes**:
  - Ensure that the formula accounts for the immediate provision of affordable homes, represented by a factor F for fast-tracking affordable housing.

Final Updated Formula

Affordable Housing Supply (AMS)
$$ \text{AMS} = (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC) + \text{Affordable Housing Stock (HS)} $$

Affordable Housing Need (AN)
$$ \text{AN} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC \times F) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)} $$

Explanation

- **D**: Diversity of housing types and tenures. Higher diversity improves absorption rates by reducing market saturation.
- **T**: Effectiveness of the direct-to-consumer model. This factor reflects how well the Home@ix dashboard and CRM system enhance sales and absorption rates.
- **F**: Fast-tracking affordable homes. This factor ensures

that affordable homes are brought forward to meet immediate demand.

Summary
By incorporating the Letwin Review's insights, the updated Home@ix Formula now better reflects the complexities of the housing market, including the importance of diverse housing types, the role of technology in improving sales, and the need to fast-track affordable housing. This comprehensive approach aims to address both supply and demand dynamics more effectively, ensuring that the affordable housing sector can meet the growing needs of the population.

You
Provide a more comprehensive analysis, showing money supply from mortgage lending, where M3 = 0.75% made up of new mortgage lending and the stock of money is a similar proportion. The money supply is also therefore tied to absorption rates and not just the supply of affordable housing through S.10 Planning Value Capture at 35%. Show that to meet affordable housing needs, effective demand has to be placed into the hands of those in affordable housing need and that this requires a new circuit of credit creation, and not an expansion of debt. The new circuit would not be inflationary and could be both rent-controlled and tied to an affordability function in line with the free housing market. Show a term which states that the free market has a maximum limit to the affordable housing provision it can make as land value capture due to the absorption rate. The free housing market and stock of mortgage debt are fundamental to the money supply due to financialisation. A new circuit of credit creation for the distinct and discreet affordable housing market has to be stated clearly and separately to make any advances in solving the problem of affordability not just in the UK but across the G7, which is a financialisation problem, not a planning or geospatial problem.

Monica

To provide a more comprehensive analysis, we need to incorporate the relationship between the money supply, mortgage lending, absorption rates, and the supply of affordable housing. We will also address the need for a new circuit of credit creation specifically for affordable housing.

Key Concepts
- **Money Supply (M3)**: Represents the total stock of money in the economy, including new mortgage lending.
- **Absorption Rates (AR)**: The rate at which new housing stock is absorbed by the market.
- **Planning Value Capture (PVC)**: The portion of value captured through planning, given as 35%.
- **Effective Demand (HD)**: The demand for affordable housing that can be met with available financial resources.
- **Affordable Housing Need (AN)**: The total need for affordable housing.
- **New Circuit of Credit Creation**: A mechanism to provide credit for affordable housing without expanding overall debt.

Analysis

Money Supply and Mortgage Lending
The money supply (M3) is

The money supply (M3) is significantly influenced by the volume of mortgage lending. Given that M3 is composed of 0.75% new mortgage lending and the stock of money is also a similar proportion, we can see how closely tied the housing market is to the overall financial system.

Money Supply from Mortgage Lending
Outstanding Value of Residential Mortgage Loans: £1,675.8 billion.
Total Value of Mortgaged Property (Pre-Potential Fall): £2.9 trillion.
Loan-to-Value Ratio (LTV): 57.786%.

Potential Value of Mortgaged Property (Post-35% Fall): £1.875 trillion.
New Mortgage Lending as Part of M3: 0.75%.
Impact of Absorption Rates
Absorption rates (AR) directly impact how quickly new housing stock, including affordable housing, is taken up by the market. A higher AR indicates that homes are sold and occupied quickly, which can stabilize or increase housing prices. Conversely, a lower AR can indicate oversupply or mismatched demand, leading to slower sales and potential price decreases.

Planning Value Capture (PVC)
Planning value capture at 35% represents the portion of the value created by new developments that can be reinvested into affordable housing projects. This mechanism can provide significant funding for affordable housing initiatives, but it alone is not sufficient to meet the entire affordable housing need.

Effective Demand and Affordable Housing Need
To meet affordable housing needs, effective demand (HD) must be placed into the hands of those in need. This requires a new circuit of credit creation that is distinct from traditional mortgage lending and does not expand overall debt levels.

New Circuit of Credit Creation
A new circuit of credit creation for affordable housing can be designed to be non-inflationary, rent-controlled, and tied to an affordability function. This circuit would operate separately from the traditional housing finance market, ensuring that funds are specifically allocated to affordable housing projects.

Incorporating the Free Market Limitations
The free market has a maximum limit to the affordable housing it can provide due to land value capture and absorption rates. The financialisation of the housing market means that the stock of mortgage debt is fundamental to the money supply.

Therefore, a new credit creation circuit for affordable housing must be clearly stated and separated from traditional market mechanisms.

Comprehensive Accounting Identity

To integrate these elements, we can refine the Home@ix Formula to include the impact of money supply, absorption rates, and the need for a new credit circuit.

Affordable Housing Supply (AMS)

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Affordable Housing Need (AN)

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HomeMaker Effective Demand (HD)

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Affordable Housing Stock (HS)

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Affordable Market Supply (AMS)

AN=HomeMaker Effective Demand (HD)+

(Housebuilding Market (HM)×P×AR×D×T×PVC×F)+Affordab...

−Affordable Market Supply (AMS)

New Circuit of Credit Creation

To represent the new circuit of credit creation (NCC), we introduce a term

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C which represents the credit specifically allocated to affordable housing without expanding overall debt.

Updated Affordable Housing Need (AN)

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Affordable Housing Stock (HS)

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Affordable Market Supply (AMS)

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New Circuit of Credit Creation (NCC)

AN=HomeMaker Effective Demand (HD)+

(Housebuilding Market (HM)×P×AR×D×T×PVC×F)+Affordab...

−Affordable Market Supply (AMS)+New Circuit of Credit Cre...

Explanation

D

D: Diversity of housing types and tenures.

T

T: Effectiveness of the direct-to-consumer model.

F

F: Fast-tracking affordable homes.

N

C

C

NCC: New Circuit of Credit Creation, representing non-inflationary, rent-controlled credit for affordable housing.

Summary

This comprehensive analysis shows that to meet affordable housing needs, effective demand must be placed into the hands of those in need through a new circuit of credit creation. This circuit should be distinct from traditional mortgage lending and should not expand overall debt levels. By incorporating the limitations of the free market and the financialisation of the housing market, we can better address the affordable housing crisis not just in the UK but across the G7. This approach recognizes that the problem is primarily financial, not merely planning or geospatial.

Monica GPT-4o

To address the comprehensive housing affordability challenge, we need to integrate the concepts of money supply, mortgage lending, absorption rates, and the need for a new circuit of credit creation into the Home@ix Formula. This approach will help us understand how financial mechanisms influence the housing market and propose solutions that are not inflationary and are specifically targeted at affordable housing.

Key Concepts

- **Money Supply (M3)**: Represents the total stock of money in the economy, including new mortgage lending.
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Analysis

Money Supply and Mortgage Lending

The money supply (M3) is significantly influenced by the volume of mortgage lending. Given that M3 is composed of 0.75% new mortgage lending and the stock of money is also a similar proportion, we can see how closely tied the housing market is to the overall financial system.

- **Outstanding Value of Residential Mortgage Loans**: £1,675.8 billion.
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- **New Mortgage Lending as Part of M3**: 0.75%.

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Planning value capture at 35% represents the portion of the value created by new developments that can be reinvested into affordable housing projects. This mechanism can provide significant funding for affordable housing initiatives, but it alone is not sufficient to meet the entire affordable housing need.

Effective Demand and Affordable Housing Need

To meet affordable housing needs, effective demand (HD) must be placed into the hands of those in need. This requires a new circuit of credit creation that is distinct from traditional mortgage lending and does not expand overall debt levels.

New Circuit of Credit Creation

A new circuit of credit creation for affordable housing can be designed to be non-inflationary, rent-controlled, and tied to an affordability function. This circuit would operate separately from the traditional housing finance market, ensuring that funds are specifically allocated to affordable housing projects.

Incorporating the Free Market Limitations

The free market has a maximum limit to the affordable housing it can provide due to land value capture and absorption rates. The financialisation of the housing market means that the stock of mortgage debt is fundamental to the money supply. Therefore, a new credit creation circuit for affordable housing must be clearly stated and separated from traditional market mechanisms.

Comprehensive Accounting Identity

To integrate these elements, we can refine the Home@ix Formula to include the impact of money supply, absorption rates, and the need for a new credit circuit.

Affordable Housing Supply (AMS)

$$\text{AMS} = (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC) + \text{Affordable Housing Stock (HS)}$$

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$$\text{AN} = \text{HomeMaker Effective Demand (HD)} + (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC \times F) + \text{Affordable Housing Stock (HS)} - \text{Affordable Market Supply (AMS)}$$

New Circuit of Credit Creation

To represent the new circuit of credit creation (NCC), we introduce a term C which represents the credit specifically allocated to affordable housing without expanding overall debt.

Updated Affordable Housing Need (AN)

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Explanation

- D: Diversity of housing types and tenures.
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Summary

This comprehensive analysis shows that to meet affordable housing needs, effective demand must be placed into the hands of those in need through a new circuit of credit creation. This circuit should be distinct from traditional mortgage lending and should not expand overall debt levels. By incorporating the limitations of the free market and the financialisation of the housing market, we can better address the affordable housing crisis not just in the UK but across the G7. This approach recognizes that the problem is primarily financial, not merely planning or geospatial.

How does the money supply (M3) influence the housing market and affordable housing stock?

Monica GPT-4o

The money supply (M3) plays a crucial role in influencing the housing market and the stock of affordable housing. M3 includes the broadest

measure of money in an economy, encompassing cash, checking deposits, savings deposits, and various other forms of near money, including large time deposits and institutional money market funds. Here's how M3 impacts the housing market and affordable housing stock:

1. **Influence on Housing Market**

Mortgage Lending

- **Availability of Credit**: An increase in M3 typically means more liquidity in the financial system, making it easier for banks to lend money. This increased availability of credit can lead to more mortgage approvals, boosting demand for housing.
- **Interest Rates**: Central banks often adjust interest rates based on the money supply. Lower interest rates reduce the cost of borrowing, encouraging more people to take out mortgages. Conversely, higher interest rates can cool down the housing market by making mortgages more expensive.
- **Housing Prices**: Increased mortgage lending can drive up housing prices as more people have access to credit and can afford to buy homes. This can lead to a housing boom. However, if the supply of housing does not keep pace with demand, it can result in significant price increases, making housing less affordable.

Investment and Speculation

- **Real Estate Investment**: When the money supply is abundant, investors may seek higher returns in the real estate market, further driving up prices. This can lead to speculative bubbles if not managed carefully.
- **Financialisation**: The financialisation of housing refers to the increasing role of financial markets and institutions in the housing sector. As more money flows into real estate, it can lead to higher prices and a focus on profit margins rather than affordability.

2. **Influence on Affordable Housing Stock**

Funding and Development

- **Government and Private Sector Funding**: An increase in M3 can provide more funding opportunities for both the government and private developers to invest in affordable housing projects. However, this requires targeted policies to ensure that the funds are directed towards affordable housing.
- **Planning Value Capture (PVC)**: As part of planning policies, a portion of the value created by new developments can be captured and reinvested into affordable housing. An abundant money supply can make it easier to implement and fund such policies.

Absorption Rates and Market Dynamics

- **Absorption Rates (AR)**: Higher absorption rates indicate that new housing stock, including affordable housing, is quickly taken up by the market. A healthy money supply can support higher absorption rates by ensuring that potential buyers and renters have access to credit.
- **Market Saturation**: If the market becomes saturated with similar types of housing, absorption rates can decline. This is where the diversity of housing types and tenures becomes crucial, as highlighted in the Letwin Review.

New Circuit of Credit Creation

- **Non-Inflationary Credit**: To effectively increase the stock of affordable housing, there needs to be a new circuit of credit creation that is specifically targeted at affordable housing. This circuit should be designed to be non-inflationary, rent-controlled, and tied to an affordability function. This ensures that the funds are used efficiently and do not contribute to overall debt expansion.

3. **Comprehensive Analysis: Money Supply and Affordable Housing**

Money Supply from Mortgage Lending

Given that M3 includes a significant portion of mortgage lending, any changes in mortgage lending practices directly affect the money supply. For example:

- **New Mortgage Lending as Part of M3**: 0.75%.

- **Impact on Housing Prices**: Increased mortgage lending can lead to higher housing prices, affecting affordability.

Affordable Housing Need (AN)

To meet affordable housing needs, effective demand must be placed into the hands of those in need. This requires:

- **Effective Demand (HD)**: Ensuring that those in need have the financial resources to access affordable housing.
- **New Circuit of Credit Creation (NCC)**: A targeted credit creation mechanism for affordable housing that does not expand overall debt levels.

Updated Home@ix Formula with Money Supply Consideration

Affordable Housing Supply (AMS)

$$\text{AMS} = (\text{Housebuilding Market (HM)} \times P \times AR \times D \times T \times PVC) + \text{Affordable Housing Stock (HS)}$$

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Summary

The money supply (M3) significantly influences the housing market by affecting mortgage lending, interest rates, and investment in real estate. These factors, in turn, impact housing prices and the availability of affordable housing. To address affordable housing needs effectively, there must be a new circuit of credit creation that is non-inflationary and specifically targeted at affordable housing. This approach ensures that funds are allocated efficiently and helps to balance the supply and demand dynamics in the housing market. By recognizing the financialisation of the housing market and the limitations of the free market in providing

affordable housing, policymakers can develop targeted strategies to improve housing affordability across the G7.

> How do fluctuations in the money supply (M3) influence the overall level of housing prices and the accessibility of affordable housing?

Monica GPT-4o

Fluctuations in the money supply (M3) have significant and multifaceted impacts on the overall level of housing prices and the accessibility of affordable housing. Here's a detailed breakdown of these influences:

1. Influence on Housing Prices

Increased Money Supply (Expansionary Phase)

- **Lower Interest Rates**: When the money supply increases, central banks often lower interest rates to encourage borrowing and spending. Lower interest rates reduce the cost of mortgages, making it cheaper for individuals to borrow money to buy homes.
- **Higher Demand for Housing**: With cheaper borrowing costs, more people can afford to take out mortgages, increasing the demand for housing. This increased demand can drive up housing prices, especially if the supply of housing does not keep pace.
- **Investment and Speculation**: An increased money supply can lead to more investment in real estate as investors seek to capitalize on rising property values. This can further inflate housing prices, particularly in markets with high speculative activity.

Decreased Money Supply (Contractionary Phase)

- **Higher Interest Rates**: When the money supply decreases, central banks may raise interest rates to control inflation. Higher interest rates increase the cost of mortgages, making it more expensive to borrow money for home purchases.
- **Reduced Demand for Housing**: Higher borrowing costs can dampen the demand for housing, as fewer people can afford to take out

mortgages. This can lead to a slowdown in housing price growth or even a decline in prices.

- **Decreased Investment**: With higher interest rates and reduced liquidity, investment in real estate may decline, leading to lower housing prices and potentially less speculative activity in the market.

2. Influence on Accessibility of Affordable Housing

Increased Money Supply

- **More Funding for Affordable Housing Projects**: An increased money supply can provide more liquidity and funding opportunities for both government and private sector initiatives aimed at developing affordable housing. This can include subsidies, grants, and low-interest loans specifically targeted at affordable housing projects.
- **Potential for Higher Prices**: While more funding can help develop affordable housing, the overall increase in housing prices due to higher demand can offset these benefits. If housing prices rise too quickly, it can make even affordable housing less accessible to lower-income individuals.
- **Inflationary Pressures**: An increased money supply can lead to inflation, which can erode the purchasing power of households. If wages do not keep pace with inflation, the affordability of housing can decline, making it harder for low-income families to afford homes.

Decreased Money Supply

- **Reduced Funding Availability**: A decrease in the money supply can lead to tighter credit conditions, making it harder for developers to secure funding for affordable housing projects. This can slow down the development of new affordable housing units.
- **Potential Stabilization of Prices**: While reduced funding can be a challenge, the potential stabilization or decline in overall housing prices due to decreased demand can make existing housing stock more affordable. However, this depends on the balance between supply and demand and the overall economic conditions.
- **Credit Constraints**: Tighter credit conditions can make it harder for low-income households to secure mortgages, even if housing prices

stabilize. This can limit accessibility to affordable housing despite potentially lower prices.

3. Comprehensive Analysis: Balancing Money Supply and Affordable Housing

Money Supply from Mortgage Lending

Given that M3 includes a significant portion of mortgage lending, fluctuations in mortgage lending practices directly affect the money supply. For example:

- **New Mortgage Lending as Part of M3**: 0.75%.
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To meet affordable housing needs, effective demand must be placed into the hands of those in need. This requires:

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$$\text{Affordable Market Supply (AMS)} + \text{New Circuit of Credit Creation (NCC)}$$

Explanation

- D: Diversity of housing types and tenures.
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- F: Fast-tracking affordable homes.
- NCC: New Circuit of Credit Creation, representing non-inflationary, rent-controlled credit for affordable housing.

Summary

Fluctuations in the money supply (M3) significantly influence housing prices and the accessibility of affordable housing. An increased money supply can lower interest rates and boost demand for housing, driving up prices and potentially making housing less affordable. Conversely, a decreased money supply can raise interest rates, reduce demand, and stabilize or lower housing prices, but it can also constrain funding for affordable housing projects and limit credit availability for low-income households.

To address these challenges, it is crucial to develop a new circuit of credit creation specifically targeted at affordable housing. This circuit should be non-inflationary, rent-controlled, and tied to an affordability function. By recognizing the financialisation of the housing market and the limitations of the free market in providing affordable housing, policymakers can develop targeted strategies to improve housing affordability across the G7. This approach ensures that funds are allocated efficiently and helps to balance the supply and demand dynamics in the housing market.

> What are the potential long-term implications of sustained increases in the money supply on the overall housing market and availability of affordable housing?

Monica GPT-4o

Sustained increases in the money supply can have profound long-term implications on the overall housing market and the availability of affordable housing. These implications can be both positive and negative, depending on how the increased money supply is managed and directed. Here's a detailed analysis:

1. **Implications for the Overall Housing Market**

Housing Prices

- **Persistent Price Inflation**: A sustained increase in the money supply often leads to persistent inflationary pressures on housing prices. With more liquidity in the economy, borrowing costs remain low, encouraging more people to take out mortgages and invest in real estate. This increased demand can drive up property prices over time.
- **Speculative Bubbles**: Continuous influxes of money into the housing market can lead to speculative bubbles, where housing prices rise significantly beyond their fundamental values. When the bubble bursts, it can lead to sharp declines in property values, financial instability, and economic recessions.

Interest Rates and Mortgage Availability

- **Prolonged Low Interest Rates**: Central banks might keep interest rates low to manage the increased money supply, which keeps mortgage rates affordable. This can encourage continuous borrowing and investment in housing, further inflating prices.
- **Credit Expansion**: With more money available, banks may loosen lending standards, leading to an increase in subprime lending. This can increase the risk of defaults and financial instability in the long run.

Market Dynamics

- **Increased Construction Activity**: Initially, a sustained increase in the money supply can stimulate construction activity as developers take advantage of cheap credit to build more housing. This can help meet demand and potentially stabilize prices if supply keeps pace.
- **Resource Allocation**: Over time, too much money chasing too few housing projects can lead to misallocation of resources, with

investments flowing into luxury or speculative projects rather than needed affordable housing.

2. **Implications for Affordable Housing**

Affordability Challenges

- **Widening Affordability Gap**: As overall housing prices rise due to sustained increases in the money supply, the affordability gap can widen. Low and moderate-income households may find it increasingly difficult to purchase homes or even afford rents, exacerbating housing inequality.
- **Increased Rental Costs**: Higher property values often translate into higher rental costs. This can make it harder for renters, particularly those with lower incomes, to find affordable housing options.

Funding and Development

- **Increased Funding Opportunities**: More liquidity in the economy can provide more opportunities for funding affordable housing projects through government initiatives, subsidies, and public-private partnerships.
- **Potential Misallocation**: Without targeted policies, the increased money supply might not be directed effectively towards affordable housing. Developers may prefer to invest in higher-margin projects rather than affordable housing, leading to a shortage in the latter.

Policy Responses

- **Increased Government Intervention**: To counteract the negative effects of sustained increases in the money supply, governments might need to intervene more aggressively in the housing market. This could include measures like rent controls, increased subsidies for affordable housing, and stricter zoning laws to ensure a proportion of new developments are affordable.
- **New Credit Mechanisms**: Developing new circuits of credit creation specifically for affordable housing can help ensure that the increased money supply benefits those in need. These mechanisms can be designed to be non-inflationary and tied to affordability criteria.

3. **Long-term Strategic Considerations**

Balanced Growth

- **Sustainable Development**: Policymakers need to ensure that the growth in the housing market is sustainable. This involves balancing the